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Course : Agile Development

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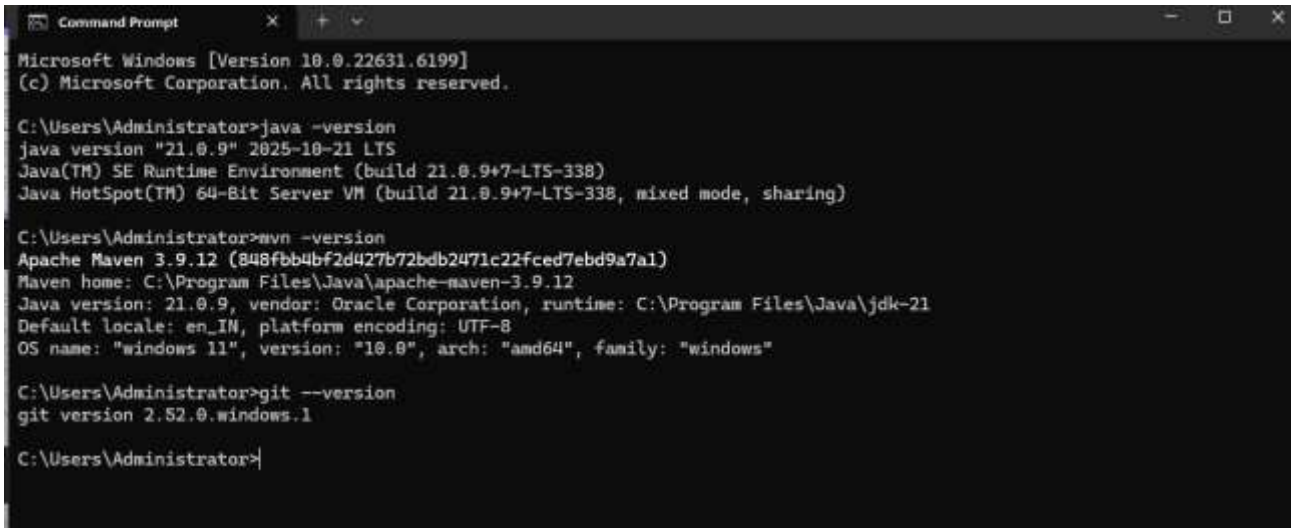
Slot: L57 + L58

Jenkins Maven Project Tutorial (Without Maven Archetype)

Step 1: Install Required Software

Install the following software: 1. Java Development Kit (JDK) 2. Apache Maven 3. Jenkins 4. Git Verify installation using:

```
java -version
mvn -version
git -version
```



Step 2: Create Maven Project Folder

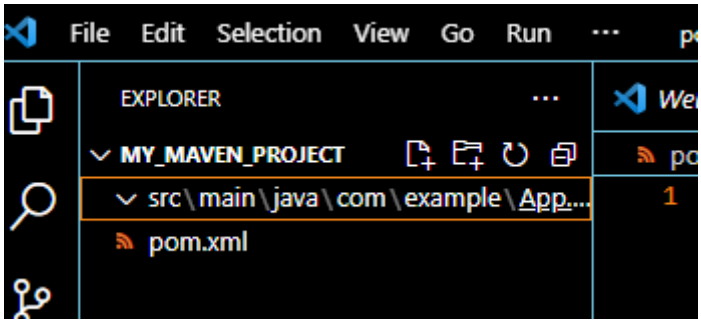
Create a new project folder manually. Example folder name:

```
my_maven_project
```

Step 3: Project Structure

Create the following directory structure inside the project folder:

```
my_maven_project
pom.xml
src
main
java
com
example
App.java
```



Step 4: Create pom.xml

Create a file named pom.xml in the root folder with the following content:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
  http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>
  <artifactId>my_maven_project</artifactId>
  <version>1.0-SNAPSHOT</version>

  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>
        <version>3.8.1</version>
        <configuration>
          <source>1.8</source>
          <target>1.8</target>
        </configuration>
      </plugin>
    </plugins>
  </build>
  </project>
```

Step 5: Create Java File

Create the file App.java in the following location:
src/main/java/com/example/App.java Add the following code:

```
package com.example; public class

App {

        public static void main(String[] args) { System.out.println("Hello Maven Jenkins Project");
        }
```

}

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.example</groupId>
 <artifactId>my_maven_project</artifactId>
 <version>1.0-SNAPSHOT</version>
 <build>
 <plugins>
 <plugin>
 <groupId>org.apache.maven.plugins</groupId>
 <artifactId>maven-compiler-plugin</artifactId>
 <version>3.8.1</version>
 <configuration>
 <source>1.8</source>
 <target>1.8</target>
 </configuration>
 </plugin>
 </plugins>
 </build>
</project>
```

Step 6: Build Maven

```
package com.example;
public class App {
 public static void main(String[] args) {
 System.out.println("Hello Maven Jenkins Project");
 }
}
```

Project Locally

Open terminal inside the project folder and run:

```
mvn clean package
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL                    powershell
PS C:\Users\Administrator\Desktop\my_maven_project> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:my_maven_project >-------------------
[INFO] Building my_maven_project 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
maven-compiler-plugin/3.8.1/maven-compiler-plugin-3.8.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/m
aven-compiler-plugin/3.8.1/maven-compiler-plugin-3.8.1.pom (12 kB at 53 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/
maven-plugins/33/maven-plugins-33.pom
```

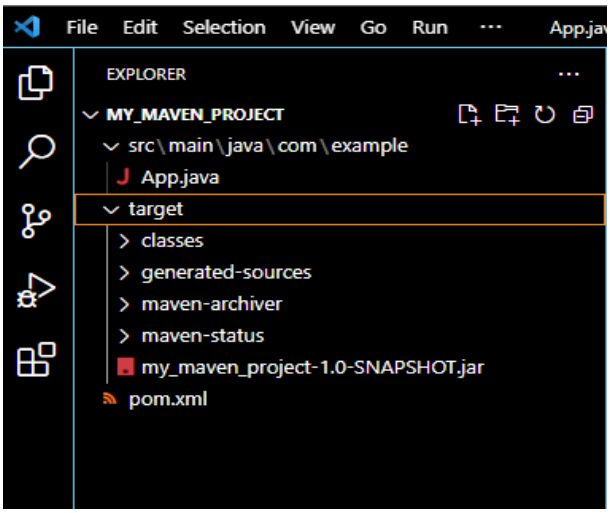
```
Downloaded from central: https://repo.maven.apache.org/maven2/org/sonatype/plexus/plexus
-sec-dispatcher/1.3/plexus-sec-dispatcher-1.3.jar (29 kB at 47 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/thoughtworks/qdox/qdo
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ my_maven_project ---
[INFO] Building jar: C:\Users\Administrator\Desktop\my_maven_project\target\my_maven_pro
ject-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.035 s
[INFO] Finished at: 2026-03-27T16:58:34+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Administrator\Desktop\my_maven_project>
```

Step 7: Build Output

If build is successful you will see:

BUILD SUCCESS

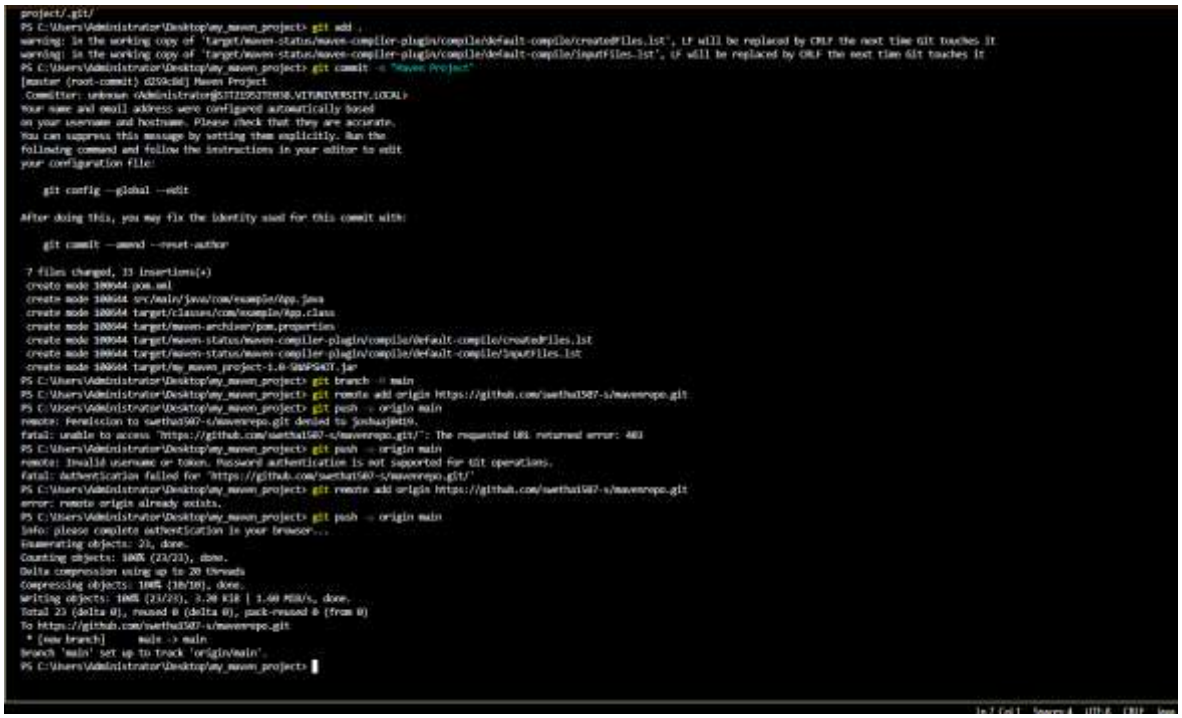
target/
my_maven_project-1.0-SNAPSHOT.jar



Step 8: Push Project to GitHub

Initialize Git repository and push the project:

```
git init git
add .
git commit -m "Maven Project" git branch -M
main
git remote add origin https://github.com/Ramlingams/my_maven_project.git git push -u origin main
```



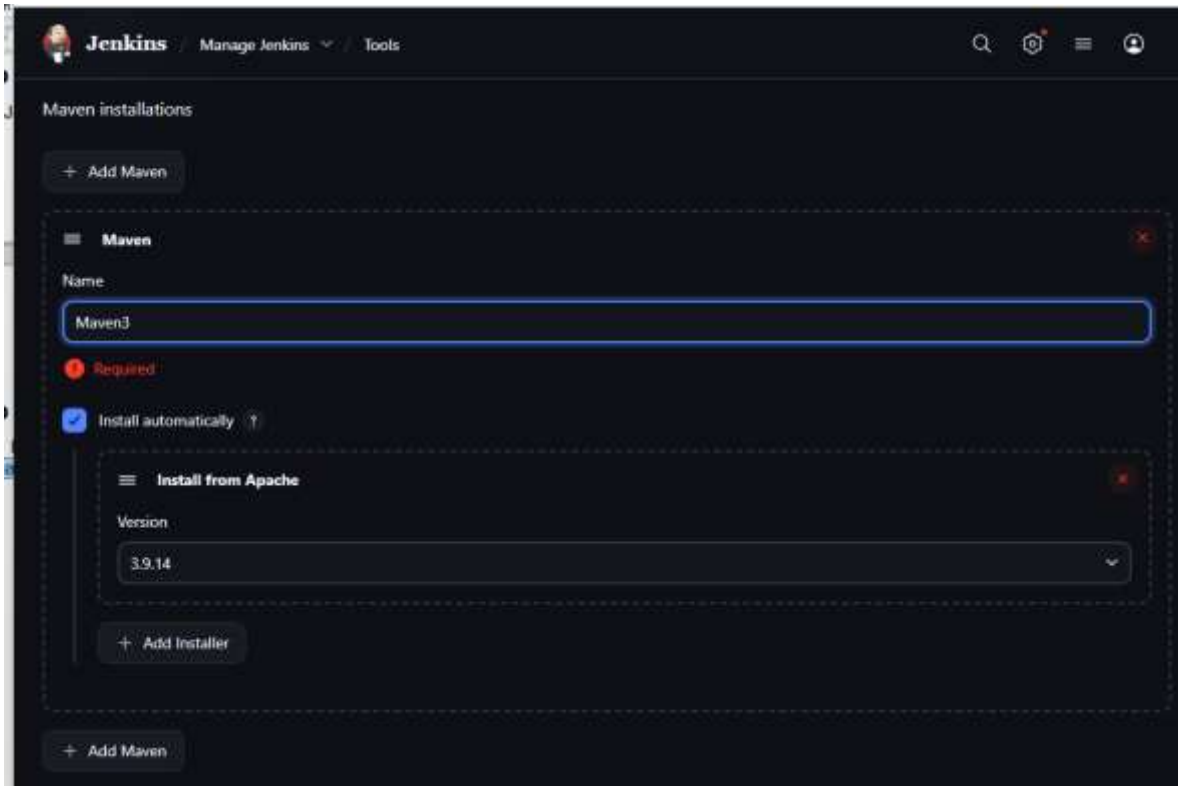
Step 9: Open Jenkins

Open Jenkins in your browser:

```
http://localhost:8080
```

Step 10: Configure Maven in Jenkins

Go to: Manage Jenkins Global Tool Configuration Add Maven installation:
Name: Maven3 Install automatically: ✓


Jenkins
Manage Jenkins
Tools
Maven installations
Add Maven
Maven
Name
Maven3
Required
Install automatically
Install from Apache
Version
3.9.14
Add Installer
Add Maven

Step 11: Create Maven Project Job

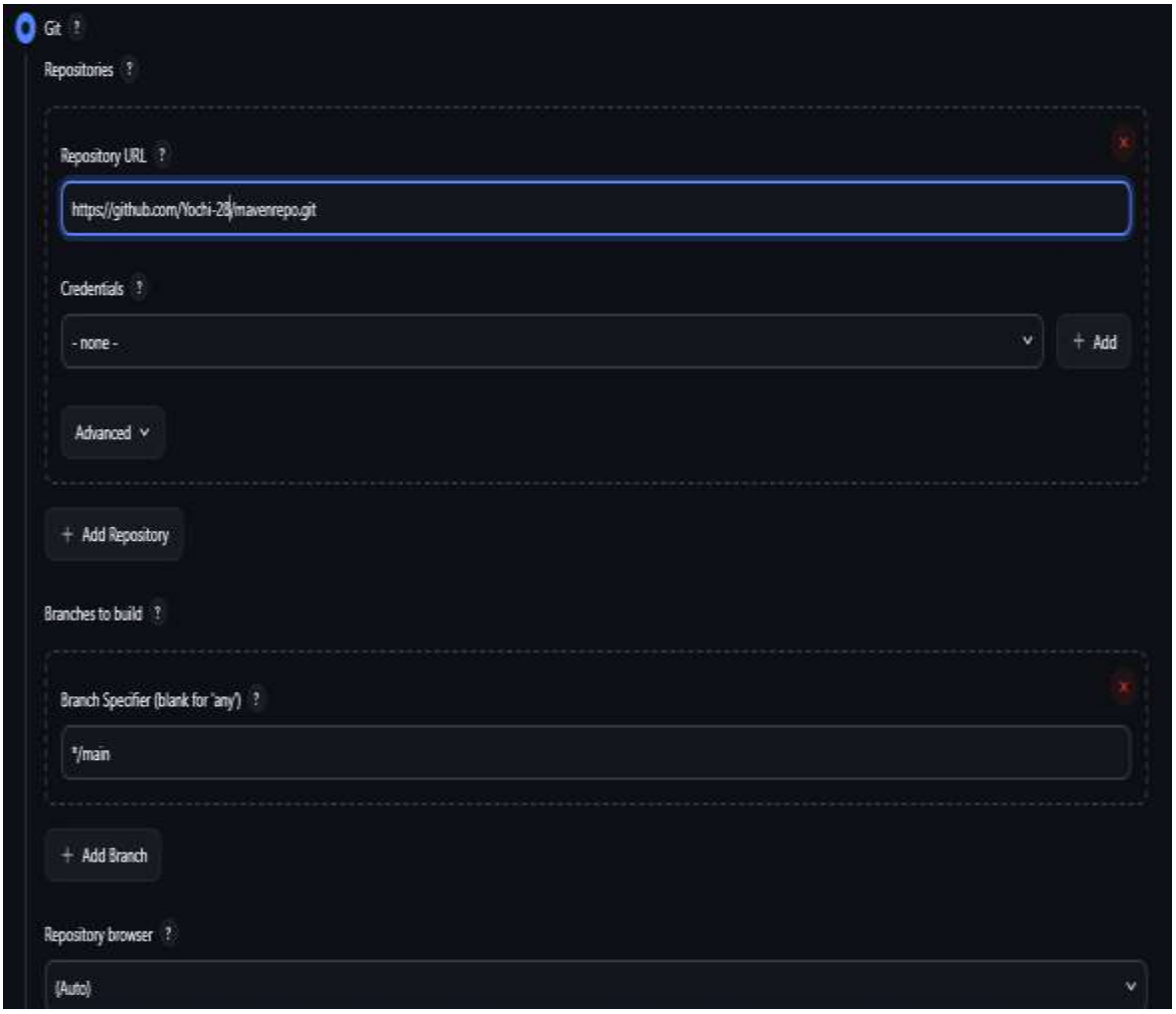
In Jenkins Dashboard: Click New Item Enter Name: MVN_01 Select: Maven Project Click OK

Step 12: Configure GitHub Repository

In Source Code Management select Git and provide:

Repository URL: https://github.com/Ramlingams/my_maven_project.git

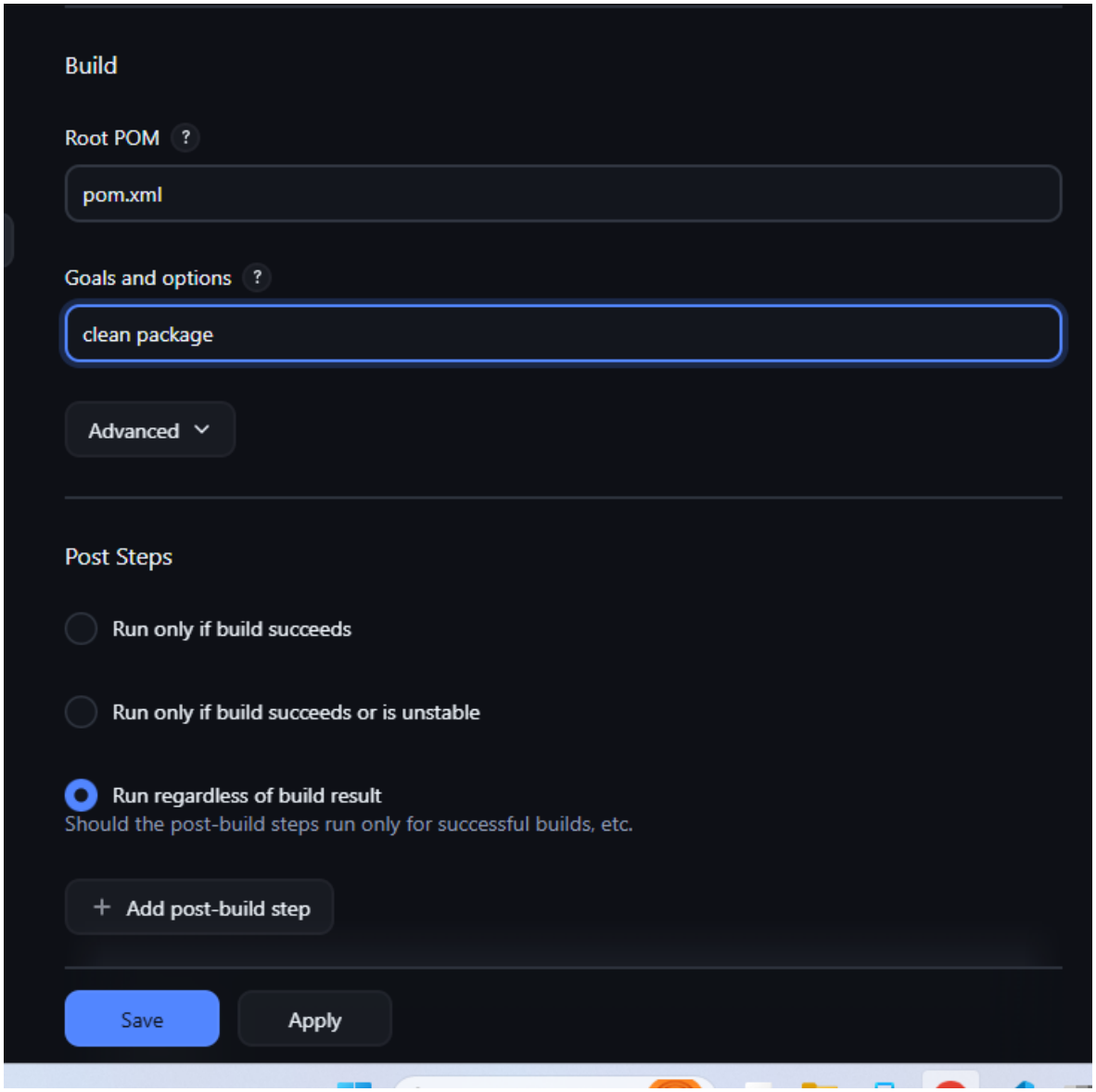
Branch:
*/main



Step 13: Configure Maven Build

In Build section provide:

Root POM : pom.xml
Goals and options : clean package



Step 14: Run Jenkins Build

Click Build Now and open Console Output. Expected result:

```
BUILD SUCCESS
```

Step 15: Final Output

The compiled JAR file will be available in Jenkins workspace:

```
workspace/MVN_01/target/my_maven_project-1.0-SNAPSHOT.jar
```

Jenkins MVN_01 #1 Console Output

```
[INFO] Downloaded from central:
https://repo.maven.apache.org/maven2/org/apache/commons/commons-
lang3/3.10.0/commons-lang3-3.10.0.jar (703 kB at 12 MB/s)
[INFO] Downloaded from central:
https://repo.maven.apache.org/maven2/org/tukaani/xz/1.10/xz-1.10.jar (169 kB at
2.3 MB/s)
[INFO] Downloaded from central:
https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.7-6/zstd-jni-
1.5.7-6.jar (7.4 MB at 46 MB/s)
[INFO] Downloaded from central:
https://repo.maven.apache.org/maven2/org/apache/commons/commons-
compress/1.20.0/commons-compress-1.20.0.jar (1.1 MB at 6.1 MB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/commons-
io/commons-io/2.21.0/commons-io-2.21.0.jar (585 kB at 2.6 MB/s)
[INFO] Building jar:
C:\ProgramData\Jenkins\.jenkins\workspace\MVN_01\target\my_maven_project-1.0-
SNAPSHOT.jar
[WARNING] Attempt to (de-)serialize anonymous class
org.jfrog.hudson.maven2.MavenDependenciesRecorder$1; see:
https://jenkins.io/redirect/serialization-of-anonymous-classes/
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  10.674 s
[INFO] Finished at: 2026-03-27T17:17:28+05:30
[INFO] ------------------------------------------------------------------------
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving C:\ProgramData\Jenkins\.jenkins\workspace\MVN_01\pom.xml to
com.example/my_maven_project/1.0-SNAPSHOT/my_maven_project-1.0-SNAPSHOT.pom
[JENKINS] Archiving
C:\ProgramData\Jenkins\.jenkins\workspace\MVN_01\target\my_maven_project-1.0-
SNAPSHOT.jar to com.example/my_maven_project/1.0-SNAPSHOT/my_maven_project-1.0-
SNAPSHOT.jar
channel stopped
Finished: SUCCESS
```